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## **Safer chairs for elderly patients: design evaluation using electromyography and force measurement**

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# **Safer chairs for elderly patients: design evaluation using electromyography and force measurements**

## **Abstract**

A vast majority of patient fall events in hospitals involve the elderly. In inpatient care settings, despite the risk of fall, patients are encouraged to leave their bed, move around their room, and sit on their chair to progress in their healing. Despite the vital role of patient chair design in improving recovery, few studies have examined the ergonomic requirements of safe patient chairs. This study examined the impact of manipulating horizontal and vertical positions of armrests in a test chair on required physical effort during Stand-to-Sit-to-Stand (St-Si-St) transitions among fifteen elderly women. Physical effort was measured using: 1) surface electromyography (sEMG); 2) force measurement by load cells; 3) video recording. Findings showed non-linear patterns of change in required physical effort due to changes in armrests' height and distance. It was also found that minimum effort is associated with armrests higher and farther apart than those in typical patient chairs.

**Keywords:** Patient chair, ergonomic seating, patient fall, healthcare furniture design, surface electromyography

**Practitioner Summary:** Safe chairs are essential for inpatient recovery, yet their ergonomic features are not investigated. Impact of changes in chair armrests on required physical effort was examined using Electromyography, force measurement, and video recording. Armrests higher and farther apart than those in typical patient chairs may be safer for elderly patients.

## **1: Introduction**

Owing to medical and scientific advances, the average projected lifespan of Americans born today is 20 years longer than those born in 1925 (Carstensen 2015). An aging society ought to provide care for older hospitalised patients who have far more complicated symptoms. They are also more vulnerable to risks like falls while staying in hospitals. In US hospitals, 3.3 to 11.5 falls happen per 1,000 patient days (Bouldin et al. 2013). 30-50 percent of falls in hospitals result in injuries with the average cost of

around \$14,000 (The Joint Commission 2015). Inside patient rooms, patient chair is one of the particularly high-risk locations for falls (Pati et al. 2017; Tzeng and Yin 2008).

Sitting on and standing from a chair can be very challenging for older patients, due to several physical limitations such as weakened muscles, poor balance, and vision problems. On the other hand, in patient rooms, encouraging patients to leave their bed, walk around their room, and sit on their chair is one way of maintaining mobility and accelerating recovery (Herman Miller 2009). However, if sitting and standing calls for considerable effort or assistance, patients may show reluctance to use their chair.

Since studies focusing specifically on patient chair design are few and scattered, ergonomic features of patient chairs have mostly followed the standards for office chairs (Herman Miller 2009). This makes sense as mechanical features of all chairs are required to follow the human anthropometric measures. However, in addition to the body size, human interaction with a piece of furniture is significantly affected by the function of furniture and user category.

This study aimed to address this gap in the field of furniture design. In clinical biomechanics and physical therapy studies sit-to-stand (Si-St) and stand-to-sit (St-Si) tests are commonly used to determine levels of physical functioning, especially in symptomatic adults. The ability of elderly patients to perform these tests may be restricted by pain, stiffness, reduced joint range of motion, muscle weakness, and incapacitation due to joint deformity (Munro et al. 1998). Janssen, Bussmann, and Stam (2002) conducted a comprehensive review of 160 studies relevant to Si-St transfer, and found 39 empirical studies with experimental design. In those experiments, among all aspects of chair design, seat height, use of armrests, and foot position were found to be the major factors affecting a successful Si-St movement and associated falls (Janssen, Bussmann, and Stam 2002). Using armrests was found to assist standing from a chair by

facilitating several mechanisms (Alexander, Schultz, and Warwick 1991; Arborelius, Wretenberg, and Lindberg 1992; Burdett et al. 1985; Schultz, Alexander, and Ashton-Miller 1992; Wheeler et al. 1985; Hu et al. 2016). Compared to other Si-St conditions (with arms free, with hands on knees, and with arms crossed), it was found that using armrests produced less average ground reaction force during the seat-off (Etnyre and Thomas 2007). The reason is the role of the armrests in supporting body weight (Gilsdorf, Patterson, and Fisher 1991). Using armrests increases the base of support (BOS) and improves stability (Bahrami et al. 2000; Schultz, Alexander, and Ashton-Miller 1992). Armrests were shown to be an aid in moving body weight forward, allowing the development of a similar horizontal force to that provided by the forward momentum of the trunk (Kralj, Jaeger, and Munih 1990; Schultz, Alexander, and Ashton-Miller 1992). Compared to Si-St, movement duration was found to be significantly longer in St-Si (Chiovetto, Patane, and Pozzo 2012; Kralj, Jaeger, and Munih 1990). This might be due to the fact that sitting down is usually performed with more precaution than standing up (Kralj, Jaeger, and Munih 1990). During the downward movement, the motor goal is twofold, including pointing a target (usually pointing the armrest with fingers) and maintaining balance. However, during the rising movement, the goal is mainly postural, that is, reaching to an upright standing position. Compared to standing up, sitting down was also found to be more affected by ageing, meaning that a higher difference between sitting and standing was observed among the elderly. Compared with younger subjects (mean age = 22.8) when sitting down, older subjects (mean age = 73.2) needed greater velocity control to keep the equilibrium in an upright initial position and to achieve accuracy before contacting the seat surface (Mourey et al. 1998).

The purpose of this study was to examine the effect of changing two key ergonomic features of a patient chair (armrests' height and distance) on the required effort to sit and stand by the elderly. The research was guided by the following research questions: (1) Does changes in armrests' height and distance have a significant effect on physical effort when being used by the elderly for sitting and standing? (2) How does the pattern of physical effort change as armrests' height and distance change? (3) What are the optimum levels of armrest height and distance to decrease the required effort for sitting and standing?

## **2: Methods**

### ***2.1. Experimental design***

A within-subject experimental study was conducted in two stages. Using a test chair, the impact of different horizontal and vertical positions of armrests on subjects' physical effort at St-Si-St transitions was examined. Electrical activity of skeletal muscles in the dominant arm and dominant leg, exerted force on the armrest by the dominant arm, and time to complete the transition were measured as indicators of physical effort. In a series of trials, first, with 9 armrest positions, it was discovered if changes in armrests' height (H) and distance (D), as well as the interaction between these two variables had a significant influence on physical effort indicators. In the second stage, based on the results of the first stage, 23 armrest positions were examined, and patterns of change in the physical effort indicators were explored. The experiments were conducted in a laboratory setting. The study protocol was approved by the Texas Tech University Human Research Protection Program, and all participants provided written informed consent.

## **2.2. Participants**

Fifteen female subjects over 65 years old (mean:  $73 \pm 5.06$ ; range: 66 to 85) participated in the study. Participants were limited to adults over 65 years of age because advanced age is shown to be a fall risk factor (Fletcher and Hirdes 2002). Older adults were also shown to adopt a similar motor strategy comparing with younger adults during the sit-to-stand transition (Papa and Cappozzo 2000). The study sample was limited to one sex to reduce variation. The reason behind selecting women was that literature suggests that women have an increased risk of falling (Fletcher and Hirdes 2002) and experiencing a fall injury (Tinetti, Doucette, Claus, and Marottoli 1995) in comparison to men. The intention was to limit the sample to those who are more vulnerable to falls and therefore more likely to be affected by changes in assistive components of their chairs.

Participants were recruited from the middle 70 percent of nationally representative range of height and weight for their sex and age group (CDC 2012). Their mean height was  $158.5 \pm 4.94$  cm (153 to 170), and their mean weight was  $71 \pm 12.35$  kg (55 to 89.5). Participants must have had the ability to conduct the required task unassisted. Other exclusion criteria were consumption of medications that affect balance or gait, history of frequent fall, and previous injuries that had affected the even use of arms or legs during daily activities. History of frequent falls was also showed to affect the adopted motor strategy for sit-to-stand (Guzmán, Porcel Melián, Cordier, & Adul Silvestre, 2010).

## **2.3. Instruments**

For a detailed description of factors considered when selecting relevant instruments, please refer to a previous publication by Valipoor and Pati (2016).

### **2.3.1. Test chair**

Valor patient chair, manufactured by Nemschoff (subsidiary of Herman Miller, Inc., Zeeland, MI), was used as the base for a test chair with adjustable armrests. An aluminium structure with telescopic and rail components was built to allow armrests' H and D adjustability from 13 to 33 centimetres and from 52 to 72 centimetres, respectively. Figure 1 shows the structure and Figure 2 shows the H and D levels within the adjustable ranges. Examined ranges were wider than the H and D ranges on available patient chairs in the market to allow for detecting possible overlooked effects.

[FIGURE 1 NEAR HERE]

[FIGURE 2 NEAR HERE]

### 2.3.2. Surface Electromyography (sEMG)

A 16-channel wireless surface EMG system (Trigno™, Delsys, Inc., Natick, MA) was used with 14g sensors (37mm x 26mm x 15mm) and an amplifier with bandwidth of 20 - 450 Hz and electrode contact area of 5mm×10mm. The sensors provided a bipolar electrode configuration with an interelectrode distance of 10 mm. The sEMG signals were digitised at a sampling rate of 1926 Hz (a pre-set, commercial hardware setting).

The sEMG sensors were attached to the biceps brachii (BB) and triceps brachii (TB) of the dominant arm (based on throwing preference), and vastus lateralis (VL) and tibialis anterior (TA) of the dominant leg (based on kicking preference). The sEMG sensors were placed over each muscle in accordance with the international guidelines put forth by the Surface EMG for Non-Invasive Assessment of Muscles (SENIAM) project (Hermens et al. 2000). sEMG recording was confined to one side only, because symmetry between dominant and non-dominant sides during walking, standing upright, Si-St, and St-Si movements has been reported in leg and arm muscles (Burnett et al.



2011; Wheeler et al. 1985). Before attaching the sensors, to assure a low skin impedance, the surface of the skin was prepared by gently shaving and cleaning the area with an alcohol wipe. The sensors were directly attached to the skin using double sided adhesive tape secured with surgical tape to minimise movement artefacts. Subjects were asked not to apply body lotion within 24 hours prior to the experiment day. Prior to testing, sEMG signal quality was examined to ensure low baseline noise and minimal line interference. Following a brief warm-up, subjects performed two isometric maximal voluntary contractions (MVCs) by each of the four muscles to determine the maximum electrical activity of each muscle when it served as an agonist. Each MVC recording was three seconds in duration. One minute of rest was allotted between MVC recordings. The mean value of two MVC recordings from each muscle was used as the reference value. Subsequent sEMG amplitude values were then normalised to those from the MVC, and expressed as a percentage.

#### *2.3.3. Load cells*

Two load cells, miniature compression load buttons by Omega Engineering Inc. (Stamford, CT), were installed under each armrest for collecting force data (data from one side was used in current study analysis). Custom-built adapters were used to digitise and time-sync the force and sEMG data together, both of which were acquired with EMGworks Acquisition software (version 4.3.1., Delsys, Inc., Natick, MA). Vertical load was shown to be the predominant load applied on the armrests by older adults specially when standing up (O'Meara and Smith 2006).

#### *2.3.4. Video cameras*

Two digital camcorders recorded the trials from two sides of the subject to be used for data synchronization and measuring duration of each run.

## **2.4. Procedure**

Each trial began with the subject standing 90 cm away from the test chair, behind a marked line on the floor. Each run was triggered by researcher's verbal command (saying 'go'). The subject then walked towards the chair and sat on it. She was asked to make herself comfortable for around ten seconds in the chair and stand up, once instructed by the researcher, and take the required steps to reach to the start point. Recordings (from sensors and video cameras) were started with the 'go' command and terminated when the subject was back on the start line. After that, adjustments were made on the armrests for the next run. The trial in each arrangement was repeated three times, with a short break taken between each two runs (while adjustments were being made) and a 5-min break when subjects were halfway through the experiment. Each subject in the first stage (4 subjects) went through 27 trial runs  $[(3H \times 3D) \times 3]$  and each subject in the second stage (11 subjects) went through 69 trial runs  $[(5H \times 5D) \times 3] - 2$ . Although adjustment to five maximum levels of H and D was possible, two runs were eliminated due to some mechanical limitations in the test chair. The mean value of the three runs of each trial was used as the representative output. The measuring sequence was randomised to minimise confounding effects related to the order of exposure.

## **2.5. Data processing and statistical analyses**

The time synced sEMG, force, and video data for each trial were analysed with EMGworks Analysis software (version 4.3.1, Delsys, Inc., Natick, MA). For each selected sEMG signal, the root-mean-square (RMS) value was used to quantify muscle activation (Basmajian and Luca 1985). Extracted RMS from two main bursts (representing the sitting and standing moments) were normalised to the RMS value from the MVC to express amplitude of the signal as a function of time. RMS was also extracted from force raw signals of two main bursts but normalised with subject weight

as a common practice in similar studies (Alexander, Schultz, and Warwick 1991; Anglin and Wyss 2000).

A two-way repeated-measures analysis of variance (ANOVA), with significance level of  $\alpha=0.05$ , was run on data from the first four subjects to determine first, whether armrest height and distance had significant effects on the electrical activity of muscles, and second, if there was an interaction between the effect of armrest height and distance. Once data was collected from all 15 subjects, the relationship between sEMG amplitude and independent variables as well as the relationship between load and independent variables were examined using polynomial regression models (linear, quadratic, and cubic). Finally, to discover if H and D associated with less effort also resulted in completing sitting and standing in a shorter time, *t*-tests and one-way ANOVA tests were used. The purpose of this stage was to assure that revealed H and D were not associated with less effort simply because these levels required more time to accomplish the tasks.

### 3. Results

Results of eight two-way repeated-measures ANOVA tests, which were run on data from the first four subjects, are shown in Table 1. These results showed that manipulating H and D significantly affected the electrical activity of almost all examined muscles (marginally significant in the BB) either in sitting or standing. The interaction between H and D also showed a significant effect on all muscles except the BB, which suggested that these two variables should not be examined separately in the rest of the experiments.

[TABLE 1 NEAR HERE]

Polynomial regression tests on data from all 15 subjects resulted in twenty line-graphs. EMG amplitude (RMS normalised to %MVC) was plotted once across H for different distances, and once across D for different heights. The same procedure was performed for load (RMS normalised to subjects' mean weight). Polynomial regression analysis was then used to determine the best model fits. These models were as follows, where  $X = H$  or  $D$ ;  $Y = \text{RMS (normalised sEMG or normalised load)}$ ; and  $B_0, B_1, B_2$ , and  $B_3 = \text{statistically determined regression coefficients}$ .

$$Y = B_0 + B_1 X \text{ (linear model) (1)}$$

$$Y = B_0 + B_1 X + B_2 X^2 \text{ (quadratic model) (2)}$$

$$Y = B_0 + B_1 X + B_2 X^2 + B_3 X^3 \text{ (cubic model) (3)}$$

For each data set, the best fit model was determined based on the calculated  $R^2$  (goodness of fit). An example of the results of polynomial regression tests is provided in Figure 3 and the parameters of regression lines in this graph are described in Table 2. In this example, as illustrated, a non-linear relationship was observed between BB sEMG amplitude and armrest distance at St-Si. Five regression lines are illustrated for five levels of H. These relationships are best fitted with the quadratic model. The goodness of fit for different H levels were calculated as: H5 ( $R^2 = 0.25$ ), H7 ( $R^2 = 0.11$ ), H9 ( $R^2 = 0.96$ ), H11 ( $R^2 = 0.77$ ), and H13 ( $R^2 = 0.14$ ). In all tests, in calculating  $R^2$ , each individual Y value was considered, instead of the mean Y value of each level of X. This strategy assured a stricter way of taking into account the variation at each measured data point in calculation of  $R^2$ , which resulted in lower  $R^2$  values in general. The result of

this test also showed that the parameters of regression lines (representing height in Table 2) were not significantly different from each other,  $F(8, 323) = 1.85$ ,  $p = 0.07$ . In other words, changes in H did not significantly affect the relationship between sEMG amplitude and D. The non-linear patterns of change in Figure 3 shows that sEMG amplitude was at its maximum level when D was adjusted to 52-57 cm, while minimum muscle activity was produced when D was 65-67 cm.

[FIGURE 3 NEAR HERE]

[TABLE 2 IMMEDIATELY AFTER FIGURE 3]

Figure 4 shows another example, a non-linear relationship between TA sEMG amplitude and armrest distance at Si-St. The relationships are best fitted with the cubic model. The parameters of regression lines to estimate the relationships are presented in Table 3. The goodness of fit for different H levels were calculated as: H5 ( $R^2 = 0.30$ ), H7 ( $R^2 = 0.38$ ), H9 ( $R^2 = 0.39$ ), H11 ( $R^2 = 0.71$ ), and H13 ( $R^2 = 0.20$ ). These non-linear patterns of change showed that sEMG amplitude was at its maximum level when D was adjusted to 67 cm, while the minimum muscle activity was produced when D was either 53-57 or 72 cm. The results of regression analysis also showed that the parameters of regression lines (representing height in Table 3) were significantly different from each other,  $F(12, 322) = 2.53$ ,  $p < 0.05$ . In other words, changes in H significantly affected the relationship between sEMG amplitude and D.

[FIGURE 4 NEAR HERE]

[TABLE 3 IMMEDIATELY AFTER FIGURE 4]

### ***3.1. Patterns of change in arm's effort generation (sitting vs. standing)***

By comparing BB and TB sEMG amplitude as well as load on the armrest in St-Si and Si-St models, it was found that five out of six regression models of the arm were cubic (with two turning points) when sitting down, while only one out of six was cubic when standing up. In other words, when sitting down, changing H and D led to higher fluctuations in the pattern of physical effort generation by arm, compared with standing up.

### ***3.2. Patterns of change in leg's effort generation (sitting vs. standing)***

Comparing with the arm models, patterns of change were not very different between St-Si and Si-St transitions in the leg muscles. At St-Si, three out of four regression models of leg were cubic, and during Si-St, all four models were cubic.

### ***3.3. Range of effort generation (sitting vs. standing)***

A comparison between St-Si and Si-St showed that the magnitude of electrical activity and force were almost in the same range in both transitions. For example, BB generated electrical activity of 8-17%MVC for sitting down and 7-11%MVC for standing up.

These ranges were also similar in TB, TA, and force amplitude. However, in VL, the range of electrical activity produced for standing up (28-45%MVC) was different from the corresponding range for sitting down (18-25%MVC).

### ***3.4. Arm muscles vs. leg muscles***

Within the examined muscles of the arm, the TB showed greater activation in both sitting and standing. The range of normalised sEMG amplitude for the TB was 14-40%MVC at St-Si and 27-45%MVC at Si-St. This range in the BB was 7-18%MVC at St-Si and 7-11%MVC at Si-St. On the contrary, the range of observed activation in both leg muscles (VL and TA) was similar in Si-St (27-45%MVC) but different in St-Si (27-

38%MVC in TB, 17-26%MVC in VL). Another difference between the two arm muscles was the level of consistency between their electrical activity and the force exerted on the armrest. Regression models showed that compared with the BB, the pattern of change in electrical activity of the TB was more similar to the pattern of change in the load on the armrests. In other words, the sEMG amplitude versus force relationship was more linear for the TB than the BB.

### **3.5. Outliers**

In all relationships, the outlier, or the model that was very different from other regression lines, either in terms of pattern or amplitude, was the result of an extreme value among the height or distance intervals. For example, in the relationship between VL sEMG amplitude and distance at Si-St, the pattern of change for H33 (highest armrest level) was totally different from the general pattern for other heights.

### **3.6. Maximum and minimum sEMG amplitude and force**

The observed maximum and minimum levels of sEMG amplitude and force (on twenty examined graphs) were integrated in a series of tables to provide a comprehensive picture of levels associated with the most and the least effort. Table 4 shows a summary of all common ranges of H and D that consistently resulted in the most and the least effort.

As shown, the maximum muscle activity and force were generated by the arm when H was adjusted to either 33cm or around 18cm and D was around 52cm. On the other hand, the minimum levels of sEMG amplitude and force were generated by the arm when H was close to 28cm and D was around 67cm. Table 4 also shows that in the leg, the maximum sEMG amplitude was generated when H was adjusted to either 13cm or close to 28cm and D was between 67 and 72 centimetres. On the other hand, the minimum level of sEMG amplitude in the leg was generated when H was close to either

18 or 33 centimetres and D was around 55cm. These results, show that the patterns of change in electrical activity of leg muscles were different from both generated sEMG amplitude and exerted force by the arm. At this stage, the levels associated with the least effort generation by the arm were selected for further examination. This decision was made based on the logic that normally, leg muscles are bigger and stronger, and hence, more relied on to perform everyday activities (Lippert 2011), except when a specific physical impairment occurs for legs. Therefore, from furniture design perspective, armrest features that lighten the effort for arms were more desirable for the majority of patient population.

[TABLE 4 NEAR HERE]

One more step was taken to determine the optimum armrests' heights and distances that ease sitting and standing by minimizing the effort required by the arm. We needed to make sure that H28 and D67 were not associated with the least effort by arms only because subjects had spent more time to accomplish the tasks. We examined the impact of the time required to accomplish each transition on H and D levels associated with maximum and minimum arm effort. The objective was to ascertain if the H and D levels that led to less electrical activity and force exertion by arms, also resulted in completing the transition in a shorter time or not, and vice versa. T-tests and one-way ANOVA tests were used to compare the total time to accomplish the transitions. The results of these tests are shown in Table 5 and Table 6. Overall, these results suggested no significant differences in terms of transition time. The only significant difference was observed between armrests' heights at St-Si,  $F = 5.60$ ,  $p < 0.05$ . A post hoc Tukey test showed that H28 and H33 differed significantly at  $p < 0.01$ . These results rejected the possibility



of the impact of transition time on the amount of effort generated by the arm at armrest levels associated with maximum and minimum arm effort, and therefore, confirmed the reliability of results in terms of H and D associated with the most and least arm effort.

[TABLE 5 NEAR HERE]

[TABLE 6 IMMEDIATELY AFTER TABLE 5]

#### **4. Discussion**

The aim of this study was to examine the impact of manipulating the horizontal and vertical positions of armrests in a test chair on physical effort during St-Si-St transitions among fifteen elderly women.

##### ***4.1. Patterns of change***

Regarding the first research question, the assumption that manipulating armrest height and distance causes a significant difference in the amount of effort required by the elderly to sit on and stand up from a patient chair was tested. Results obtained from two series of data analysis confirmed this assumption. ANOVA results from the first four subjects showed that electrical activity of almost all examined muscles (triceps, VL, TA, and marginally BB) significantly increased or decreased as a result of changes in H or D or a cumulative effect of both. With reference to the second research question, after increasing the number of subjects and range of heights and distances, regression models indicated that 1) the change in physical effort due to a change in H and D does follow specific patterns and 2) these identified patterns are almost always nonlinear and can be described by cubic and quadratic regression models.

A qualitative comparison between regression models in the arm and the leg showed that the patterns of physical effort generation in the arm were steadier (with less

fluctuations) at Si-St, compared with St-Sit, while these patterns were not very different between St-Si and Si-St transitions in the leg. This result should not be surprising as it has been shown in published literature that sitting down is a more complicated action compared to standing up (Kralj, Jaeger, and Munih 1990). This complication could be due to higher number of motor goals and difficulty of maintaining balance while reaching a target while sitting. What the current study results highlights is a higher impact of this challenge on arms, compared with legs. The leg's goal is more straightforward, which is maintaining the balance. In addition, in the published literature, sitting down has been shown to be more affected by ageing. It was found that compared to a younger group, older subjects needed greater velocity control to keep the equilibrium in an upright initial position before contacting the seat surface (Mourey et al. 1998). Therefore, what we have observed in terms of more complexity of sitting down for arms compared to legs, is probably more specific to the elderly.

Another comparison showed that it took the same range of effort to sit and stand for all muscles except for VL. Considering the function of VL, this result makes sense, because this muscle is a knee extensor (Lippert 2011), which makes it more effective in extending the knee for standing up than flexing it for sitting down. The higher electrical activity by VL at Si-St compared with St-Si was also observed by Wheeler et al. (1985) who reported a greater activity by VL among older people (with mean age of 75) than a younger group (with mean age of 24) when standing up. Finally, a comparison between arm muscles showed that the electrical activity of the TB was higher than BB both in sitting and standing. It was also more similar to the pattern of load exerted on the armrests. This result suggests that the TB is more involved in using the armrests and should receive a higher attention in future studies, especially when there is a limitation in the number of muscles under examination.

Comparing regression models of arm and leg showed that generally, the patterns of change in electrical activity of leg muscles were in the opposite direction of the change in both arm sEMG amplitude and exerted force. In other words, the H and D that resulted in the lowest amplitude of arm sEMG and force were associated with the highest leg sEMG amplitude at the same phase (either in sitting or standing).

Considering Si-St, this result might indicate the underlying mechanism for the prevalent strategy among the elderly to use their upper extremities to improve their stand-up performance (Bohannon and Corrigan 2003). From chair design perspective, this balance could also mean that if a chair is designed in a way that calls for higher effort by the legs, the required effort by the arms will be decreased. The benefit of this trade-off might be intuitively justifiable, but to scientifically support it, examining a higher number of muscle groups in arms and legs in future studies will be very beneficial. If we know that a specific ergonomic feature (e.g. distance between arms) decreases the required effort by arms but increases the required effort by legs, we will be able to recommend a specific range of design features (e.g. armrest location) for patients with upper limb use restrictions. For instance, some elderly patients suffer from shoulder pain due to pathologies of tendon of the long head of the biceps (LHB) (Longo et al. 2011). In such case, patients are recommended to avoid overusing their arm during recovery as much as possible.

#### **4.2. Optimum Levels**

With regards to the third research question, regression models showed that the least sEMG amplitude and force by the arm were around H of 28cm and D of 66-69cm. One exception was the lowest level of BB sEMG amplitude versus height at Si-St, which occurred at not one but two H ranges (13-14cm and 25-28cm). Since the higher interval (25-28cm) was in the same range as other armrests' heights associated with the least

sEMG amplitude and force, we concluded that the common range of height that consistently resulted in lower effort was around H11. Patient chairs in the market have armrests of 7-25cm high. Previous studies had also suggested 25cm for the armrests on chairs for the elderly (Finlay et al. 1983; Wheeler et al. 1985) and for the patients (Hanger, Ball, and Mulley 1991). To determine the optimum levels, the time required to accomplish each transition on H and D associated with maximum and minimum arm effort were also compared. The result showed no significant difference in transition times, but the transitions on H and D associated with minimum effort were consistently performed in a shorter time.

The distance between two armrests was not specifically studied before, but the chairs in the market has armrests with 56-66 internal distance. This means that the range of armrest H and D in available patient armchairs are typically lower and smaller than the optimal values identified in this study. Results also showed that the peak sEMG amplitude and force by arm occurred in the highest (H33) and near the lowest levels of armrests (H15-18), as well as the closest distance between them (H52). This was perhaps due to delimiting our sample to the middle 70 percent of nationally representative range of height and weight for women over 65 years of age.

#### **4.3. Limitations**

In this study, although there was more control over extraneous variables that may bias the results in a natural setting, the generalizability of results was reduced owing to the contrived setting, as well as possible atypical behaviour of subjects. Surface EMG sensors were attached to subjects' arms and legs, and they were aware of study objectives. Therefore, the question remains pertaining to differences in the world beyond the laboratory. Another limitation was about subjects' attributes which was from a restricted range closer to the average. Selecting a homogeneous group, helped us

to decrease the variations in data, but limited the generalizability of results to the broader population. Although no functional scale was used, the heterogeneity among participants was raised through the described exclusion criteria. It is important to note that despite our effort to exclude subjects in the top and bottom 15 percent of the height and weight range for their sex and age group, the difference existed in participants' arms, legs, and trunk length, and may have affected the results.

The generalizability of this study is also limited by the sample size. Conducting research with fifteen subjects limited the application of findings to the larger population, but allowed for an in-depth study of the phenomenon. Since this study found that relationships between armrest attributes and physical efforts exist, and the adopted techniques have demonstrated the potential for meaningful examination, future studies may extend the current effort to a larger and broader sample and more number of variables. Finally, although the effect of fatigue was controlled in this study by randomised-trial design and providing breaks for subjects, there is always a possibility that subjects may have fatigued due to the large number of trials.

## **5. Conclusion**

The ability to sit on and stand up from a chair without assistance is considered a benchmark for good clinical recovery, particularly in older patients. While for a young and healthy person, sitting and standing may be performed with ease and little need for physical support, for the elderly or those with a history of injury or pathology, these manoeuvres may present considerable difficulty. This study took a step towards reducing this difficulty by examining one of the crucial ergonomic attributes of a patient chair. Being exploratory in nature, this study raises several critical questions for future studies -some from a design perspective, and some from a methodological perspective. In terms of design, the results of this study provide new insights on designing and

evaluating patient chairs. We found that chairs with higher and farther armrests than those in the market may facilitate a safer St-Si-St transition. A significant discovery was that armrest features that ease sitting and standing for the arms, may require a higher effort by the legs. In terms of methods, a consistent change was observed in effort indicators as a result of change in armrest features, which is an indicator of the ability of applied techniques to detect meaningful changes. Therefore, the same techniques may be used with more variables and subjects to confirm the results reported here and discover more about optimal seating solutions.

In the future, other techniques may be integrated to reinforce the applied method in this study, such as ground reaction force measurement. This will increase the number of effort indicators from upper and lower limbs. With the same rationale, a more number of muscles may be examined using sEMG, such as muscles of forearm. Four examined muscles in this study were selected since more attention was accorded to them in previous studies. However, the current study was one of the initial attempts to measure changes in muscle sEMG amplitude as a result of changes in patient chair attributes. Similar studies on other muscles of arms and legs may contribute to a better understanding of the most engaged muscles in using armrests. Furthermore, extending the experiments to muscles in trunk, hip, or shoulders will take into consideration other body segments that may play an important role in sitting and standing, and therefore affected by different design of armrests. This study showed that a chair armrest that lighten the effort of arms when sitting and standing may increase the effort of legs. This finding may be examined further to determine if patients with upper-limb dysfunctions should use chairs that should be avoided by those with lower-limb dysfunctions.

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Table 1. ANOVA results. ANOVA tests examined the effects of H, D, and the interaction between H and D on sEMG amplitude of the four engaged muscles in the four first subjects.

| Dep V | Indp V | Task  | Muscle | (DF <sub>n</sub> , DF <sub>d</sub> ) | F     | p       |
|-------|--------|-------|--------|--------------------------------------|-------|---------|
| EMG   | H      | St-Si | BB     | (2, 6)                               | 4.64  | 0.060   |
| EMG   | D      | St-Si | BB     | (2, 6)                               | 1.46  | 0.304   |
| EMG   | H×D    | St-Si | BB     | (4, 12)                              | 1.59  | 0.239   |
| EMG   | H      | Si-St | BB     | (2, 6)                               | 0.70  | 0.531   |
| EMG   | D      | Si-St | BB     | (2, 6)                               | 1.75  | 0.253   |
| EMG   | H×D    | Si-St | BB     | (4, 12)                              | 1.95  | 0.166   |
| EMG   | H      | St-Si | TB     | (2, 6)                               | 7.30  | 0.025*  |
| EMG   | D      | St-Si | TB     | (2, 6)                               | 7.61  | 0.023*  |
| EMG   | H×D    | St-Si | TB     | (4, 12)                              | 3.61  | 0.037*  |
| EMG   | H      | Si-St | TB     | (2, 6)                               | 5.68  | 0.041*  |
| EMG   | D      | Si-St | TB     | (2, 6)                               | 9.80  | 0.013*  |
| EMG   | H×D    | Si-St | TB     | (4, 12)                              | 0.67  | 0.627   |
| EMG   | H      | St-Si | VL     | (2, 6)                               | 5.01  | 0.053   |
| EMG   | D      | St-Si | VL     | (2, 6)                               | 6.74  | 0.029*  |
| EMG   | H×D    | St-Si | VL     | (4, 12)                              | 0.58  | 0.680   |
| EMG   | H      | Si-St | VL     | (2, 6)                               | 15.93 | 0.004** |
| EMG   | D      | Si-St | VL     | (2, 6)                               | 11.76 | 0.008** |
| EMG   | H×D    | Si-St | VL     | (4, 12)                              | 3.42  | 0.043*  |
| EMG   | H      | St-Si | TA     | (2, 6)                               | 0.24  | 0.790   |
| EMG   | D      | St-Si | TA     | (2, 6)                               | 10.60 | 0.011*  |
| EMG   | H×D    | St-Si | TA     | (4, 12)                              | 6.24  | 0.006** |
| EMG   | H      | Si-St | TA     | (2, 6)                               | 16.99 | 0.003** |
| EMG   | D      | Si-St | TA     | (2, 6)                               | 3.17  | 0.115   |
| EMG   | H×D    | Si-St | TA     | (4, 12)                              | 0.64  | 0.645   |

\* p < 0.05, \*\* p < 0.01

Table 2. BB sEMG vs. Distance at St-Si. Parameters of regression lines (best fit values)

|                     | H=13             | H=18             | H=23             | H=28             | H=33             |
|---------------------|------------------|------------------|------------------|------------------|------------------|
| B <sub>0</sub> (SD) | 0.1321 (0.0082)  | 0.1197 (0.0080)  | 0.0885 (0.0041)  | 0.0944 (0.0057)  | 0.1395 (0.0060)  |
| B <sub>1</sub> (SD) | -0.0054 (0.0023) | -0.0017 (0.0023) | -0.0058 (0.0009) | -0.0073 (0.0013) | -0.0020 (0.0014) |
| B <sub>2</sub> (SD) | 0.0025 (0.0013)  | 0.0022 (0.0013)  | 0.0026 (0.0004)  | 0.0024 (0.0005)  | 0.0011 (0.0006)  |

Table 3. TA sEMG vs. Distance at Si-St. Parameters of regression lines (best fit values)

|                     | H=13             | H=18             | H=23             | H=28             | H=33             |
|---------------------|------------------|------------------|------------------|------------------|------------------|
| B <sub>0</sub> (SD) | 0.4099 (0.0114)  | 0.3587 (0.0117)  | 0.3974 (0.0102)  | 0.368 (0.0046)   | 0.3284 (0.0088)  |
| B <sub>1</sub> (SD) | 0.0081 (0.0114)  | 0.0302 (0.0116)  | 0.0235 (0.007)   | 0.0207 (0.0031)  | 0.0117 (0.006)   |
| B <sub>2</sub> (SD) | -0.0055 (0.0017) | -0.0051 (0.0018) | -0.0024 (0.001)  | -0.0002 (0.0004) | -0.0016 (0.0008) |
| B <sub>3</sub> (SD) | -0.0009 (0.0013) | -0.0031 (0.0014) | -0.0013 (0.0005) | -0.0012 (0.0002) | -0.0009 (0.0004) |

Table 4. Levels of H and D associated with maximum and minimum EMG and force in the arm and the leg.

| Limb | Variable | Effort | Size (cm) |
|------|----------|--------|-----------|
| Arm  | Height   | Max    | 13 ; 33   |
|      |          | Min    | 28        |
|      | Distance | Max    | 52        |
|      |          | Min    | 67        |
| Leg  | Height   | Max    | 13 ; 28   |
|      |          | Min    | 18 ; 33   |
|      | Distance | Max    | 63.5 <    |
|      |          | Min    | 53 - 58   |

Table 5. T-test Results (transition times at D levels associated with Max and Min arm effort)

|       | D <sub>Max-Effort</sub> 52 |       | D <sub>Min-Effort</sub> 67 |       | t     | P value |
|-------|----------------------------|-------|----------------------------|-------|-------|---------|
|       | Mean                       | SEM   | Mean                       | SEM   |       |         |
| St-Si | 2.067                      | 0.058 | 1.959                      | 0.052 | 1.326 | 0.1908  |
| Si-St | 1.455                      | 0.041 | 1.378                      | 0.036 | 1.353 | 0.1821  |

Table 6. ANOVA Results (transition times at H levels associated with Max and Min arm effort)

| Transition | Levels             | ANOVA statistics | SS    | MS    | F     | P value |
|------------|--------------------|------------------|-------|-------|-------|---------|
| Si-St      | H7 vs. H28 vs. H33 | Between columns  | 0.241 | 0.120 | 2.483 | 0.090   |
|            |                    | Within columns   | 3.780 | 0.048 |       |         |
|            |                    | Total            | 4.020 |       |       |         |
| Si-St      | H7 vs. H28 vs. H33 | Between columns  | 0.604 | 0.302 | 5.597 | 0.005** |
|            |                    | Within columns   | 4.210 | 0.054 |       |         |
|            |                    | Total            | 4.814 |       |       |         |

\*\* p<0.01







